

3. Project Design Phase

- The Project Design Phase converts the requirements into a structured technical design. In the context of importing spreadsheet data using Transform Maps, this phase defines how data will flow from the spreadsheet to the final target table.

The basic architecture can be represented as:

Spreadsheet → Import/Source Table → Transform Map → Target Table

The spreadsheet acts as the source of the data. When the spreadsheet is uploaded, its records are stored in an intermediate import table. The Transform Map then reads the source records and determines how they should be processed. Finally, the transformed records are inserted into or updated in the target table.

- One of the most important design activities is field mapping. A field-mapping document should be prepared before creating the Transform Map. For **example**:

Spreadsheet Field

Source Field

Target Field

Employee ID

employee_id

Employee ID

Employee Name

employee_name

Name

Email

email

Email

Department

department

Department

Location

location

Location

Joining Date

joining_date

Joining Date

- The design must also determine which field should act as the unique identifier. For example,

Employee ID can be selected as a coalesce field if it uniquely identifies every employee. This helps the system decide whether an incoming row represents an existing record or a new record.

The design should also define transformation rules. For example, the spreadsheet may contain department names in different formats such as "IT Dept", "Information Technology", and "IT". A transformation rule can standardize these values before they are stored in the target table.

Error handling is another part of the design. If a source record contains an invalid value, the system should identify the error instead of silently importing incorrect information. The design may specify that invalid records are rejected, skipped, or logged for correction.

- The project design should also define the testing environment. Development or test data should be used before importing actual organizational information. A backup or recovery approach should be considered when importing important data.

The final design should therefore describe the complete data flow, table structure, field mapping, transformation logic, validation rules, duplicate handling, security requirements, and error-handling mechanism.

- A well-prepared design provides developers with a clear blueprint for implementing the Transform Map and reduces confusion during development.